Physics 9th – Chapter-6-Work and Energy (Short Response Questions)

Q1. A car is moving with a constant speed along a straight road. Is there any work done on the car?

Ans:

- The engine force does positive work, but friction and air resistance do an equal amount of negative work.
- The net force on the car is zero, so the net work done on the car is zero.
- Concept: by the work-energy theorem, net work = change in kinetic energy, and the car's kinetic energy stays constant.

Q2. Does the work done in raising a box up in a building depend upon how fast it is raised up? Through which path? To how much height?

Ans:

- No. Work done = mgh, which depends only on the weight of the box and the height raised.
- It does not depend on how fast it is raised or on the path (stairs or lift); raising it faster only needs more power.
- It depends on the height: the greater the height, the greater the work done (and the potential energy gained).

Q3. Work done on the body either speeds it up or slows it down. Keeping it in mind, explain how much work is done by centripetal force on an orbiting satellite?

Ans:

- The centripetal force on a satellite acts towards the centre, perpendicular to its velocity (displacement).
- Work $W = Fd \cos 90° = 0$, so the centripetal force does no work.
- That is why it changes only the direction of motion; the speed and kinetic energy of the satellite remain constant.

Q4. A car has kinetic energy E_k. By what factor its kinetic energy would change, if its velocity is doubled?

Ans:

- $E_k = \frac{1}{2}mv^2$, so kinetic energy is proportional to the square of velocity.
- If v becomes 2v, then $E_k' = \frac{1}{2}m(2v)^2 = 4 \times \frac{1}{2}mv^2 = 4E_k$.
- So the kinetic energy becomes four times.

Q5. A bullet is fired from gun, bullet penetrates into sand wall and it stops. Where does its kinetic energy used?

Ans:

- The kinetic energy of the bullet is used in doing work against the resistive force of the sand.
- It is converted mainly into heat (bullet and sand get warm), some sound energy, and in pushing and deforming the sand.
- Concept: energy is not destroyed; it changes form (law of conservation of energy).

Q6. An LED light bulb has efficiency of 80%. Does it violate conservation of energy principle?

Ans:

- No. 80% of the electrical energy is converted into light (useful output).
- The remaining 20% is converted into heat (wasted energy), not destroyed.
- Input energy = useful output + wasted energy, so total energy remains conserved.

Q7. How does using renewable energy sources contribute to reducing environmental impact compared to non-renewable sources?

Ans:

- Renewable sources (solar, wind, hydro, tidal) produce little or no greenhouse gases and air pollutants during operation.
- Fossil fuels emit CO_2 and harmful gases that cause air pollution, acid rain and climate change; nuclear fuel produces radioactive waste.
- Renewables are naturally replenished, so they reduce mining, pollution and dependence on limited fuels.

Q8. Will we eventually rely entirely on renewable energy sources? Why or why not?

Ans:

- Not completely in the near future: solar and wind are intermittent (depend on weather and time of day) and need energy storage.
- Renewables also need large land areas, high initial cost and suitable locations.
- However, as technology and storage improve and fossil fuels run out, our dependence on renewables will keep increasing.

Q9. How can increasing the power of a machine impact its energy consumption?

Ans:

- Power is the rate of using energy: $E = P \times t$.
- A more powerful machine does the same work in less time, but it uses energy at a faster rate.
- If it runs for the same time, it consumes more energy (more units, kWh), e.g. a 2 kW heater uses double the energy of a 1 kW heater per hour.

Q10. A perpetual engine has an efficiency equal to 1. Why it will not work?

Ans:

- Efficiency 1 (100%) means no energy is wasted, but friction and heat losses are always present in real machines.
- A perpetual motion machine would work forever without any external energy source, which violates the law of conservation of energy.
- Hence no practical engine can have efficiency equal to 1, so a perpetual engine is impossible.